

Name: _____ Date: _____ Period: _____

1. A quadratic model gives the height of a projectile in feet: $f(x) = -2x^2 + 12x$, where x is time in seconds.

(a) Evaluate $f(3)$ and $f(5)$.

$$f(3) = -2(\underline{\hspace{1cm}})^2 + 12(\underline{\hspace{1cm}}) = \underline{\hspace{2cm}} \text{ ft}$$

$$f(5) = -2(\underline{\hspace{1cm}})^2 + 12(\underline{\hspace{1cm}}) = \underline{\hspace{2cm}} \text{ ft}$$

(b) Compute the average rate of change of f on $[3, 5]$. Include units.

$$\text{ARC} = \frac{f(5) - f(3)}{5 - 3} = \frac{\underline{\hspace{2cm}}}{\underline{\hspace{1cm}}} = \underline{\hspace{2cm}}$$

Units: _____

2. A rational model for the force between two magnets is $P(d) = \frac{50}{d^2}$, where d is the distance in meters between the magnets.

State the domain restriction and explain in one sentence why it applies.

Domain: _____ Explanation: _____

3. The average rate of change of the pen area $A(w)$ on $[6, 8]$ is $-4 \text{ ft}^2/\text{ft}$.

What does this tell you about the pen design on that interval? Answer in a complete sentence.

